

PYTHON MINI PROJECT- CUSTOMER BEHAVIOR ANALYSIS

1) VS-CODE FOR DATA LOADING AND EXPLORATION

```
import pandas as pd
from sqlalchemy import create_engine
username="postgres"
password="ira@26082006"
host="localhost"
port="5432"
database="customer_behavior"
password=password.replace("@","%40")

df=pd.read_excel(r"C:\Users\Ira\Downloads\customer_shopping_behavior.xlsx"
)

print(df.head())

print(df.info())
print(df.describe(include='all'))
print(df.isnull().sum())
df['Review Rating']=df.groupby('Category')['Review
Rating'].transform(lambda x:x.fillna(x.median()))
print(df.isnull().sum())
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_')
df=df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
print(df.columns)
# Create a column age_group
labels=['Young Adult','Adult','Middle-Aged','Senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
print(df[['age','age_group']].head(10))
# Create column purchase_frequency_days
frequency_mapping={
    'Fortnightly':14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
```

```

        'Annually': 365,
        'Every 3 Months': 90
    }
    df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)
    print(df[['purchase_frequency_days','frequency_of_purchases']].head(10))
    print(df['frequency_of_purchases'].unique())
    df=df.drop('promo_code_used',axis=1)
    print(df.columns)

# Connect to Postgresql
engine=create_engine("postgresql+psycopg2://postgres:ira%4026082006@localhost:5432/customer_behavior")
#Load dataframe into postgresql
table_name="customer"
df.to_sql('customer',engine,if_exists="replace",index=False)
print(f>Data successfully loaded into table 'customer' in database 'customer_behavior'.")

```

1) df.head()

PS C:\Users\Ira\Downloads> python project.py

	Customer ID	Age	Gender	Item Purchased	Category	Discount Applied	Promo Code
Used Previous Purchases	0	1	55	Male	Blouse Clothing ...	Yes	Yes
Payment Method	1	2	19	Male	Sweater Clothing ...	Yes	Yes
Frequency of Purchases	2	3	50	Male	Jeans Clothing ...	Yes	Yes
	3	4	21	Male	Sandals Footwear ...	Yes	Yes
	4	5	45	Male	Blouse Clothing ...	Yes	Yes

2) df.describe()

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\Ira\Downloads> &

C:\Users\Ira\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/Ira/Downloads/project.py

	Customer ID	Age	Gender	Item Purchased	Category	Discount Applied	Promo Code
0	1	55	Male	Blouse	Clothing ...	Yes	Yes
1	2	19	Male	Sweater	Clothing ...	Yes	Yes
2	3	50	Male	Jeans	Clothing ...	Yes	Yes
3	4	21	Male	Sandals	Footwear ...	Yes	Yes
4	5	45	Male	Blouse	Clothing ...	Yes	Yes

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	str
3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads> &

C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/lra/Downloads/project.py

	Customer ID	Age	Gender	Item Purchased	Category	...	Discount Applied	Promo Code	
0	1	55	Male	Blouse	Clothing	...	Yes	Yes	14
Venmo									
		Fortnightly							
1	2	19	Male	Sweater	Clothing	...	Yes	Yes	2
Cash									
		Fortnightly							
2	3	50	Male	Jeans	Clothing	...	Yes	Yes	23
Card									Credit
		Weekly							
3	4	21	Male	Sandals	Footwear	...	Yes	Yes	49
PayPal									
		Weekly							
4	5	45	Male	Blouse	Clothing	...	Yes	Yes	31
PayPal									
		Annually							

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
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3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
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12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

	Customer ID	Age	Gender	Item Purchased	...	Promo Code Used	Previous Purchases
count	3900.000000	3900.000000	3900	3900	...	3900	3900.000000
3900		3900					

unique	NaN	NaN	2	25 ...	2	NaN	6
7							
top	NaN	NaN	Male	Blouse ...	No	NaN	PayPal
Every 3 Months							
freq	NaN	NaN	2652	171 ...	2223	NaN	677
584							
mean	1950.500000	44.068462	NaN	NaN ...	NaN	25.351538	
NaN	NaN						
std	1125.977353	15.207589	NaN	NaN ...	NaN	14.447125	
NaN	NaN						
min	1.000000	18.000000	NaN	NaN ...	NaN	1.000000	
NaN	NaN						
25%	975.750000	31.000000	NaN	NaN ...	NaN	13.000000	
NaN	NaN						
50%	1950.500000	44.000000	NaN	NaN ...	NaN	25.000000	
NaN	NaN						
75%	2925.250000	57.000000	NaN	NaN ...	NaN	38.000000	
NaN	NaN						
max	3900.000000	70.000000	NaN	NaN ...	NaN	50.000000	
NaN	NaN						

[11 rows x 18 columns]

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads> &

C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/lra/Downloads/project.py

	Customer ID	Age	Gender	Item Purchased	Category ...	Discount Applied	Promo Code
Used Previous Purchases							
Payment Method							
Frequency of Purchases							
0	1	55	Male	Blouse Clothing ...	Yes	Yes	14
Venmo							
		Fortnightly					
1	2	19	Male	Sweater Clothing ...	Yes	Yes	2
Cash							
		Fortnightly					
2	3	50	Male	Jeans Clothing ...	Yes	Yes	23
Credit Card							
		Weekly					
3	4	21	Male	Sandals Footwear ...	Yes	Yes	49
PayPal							
		Weekly					
4	5	45	Male	Blouse Clothing ...	Yes	Yes	31
PayPal							
		Annually					

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
---	--------	----------------	-------

```

--- -----
0  Customer ID      3900 non-null  int64
1  Age              3900 non-null  int64
2  Gender           3900 non-null  str
3  Item Purchased   3900 non-null  str
4  Category         3900 non-null  str
5  Purchase Amount (USD) 3900 non-null  int64
6  Location         3900 non-null  str
7  Size             3900 non-null  str
8  Color            3900 non-null  str
9  Season           3900 non-null  str
10 Review Rating    3863 non-null  float64
11 Subscription Status 3900 non-null  str
12 Shipping Type    3900 non-null  str
13 Discount Applied 3900 non-null  str
14 Promo Code Used  3900 non-null  str
15 Previous Purchases 3900 non-null  int64
16 Payment Method   3900 non-null  str
17 Frequency of Purchases 3900 non-null  str

```

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

```

Customer ID      Age Gender Item Purchased ... Promo Code Used Previous
Purchases Payment Method Frequency of Purchases
count  3900.000000 3900.000000 3900      3900 ...      3900      3900.000000
3900      3900
unique      NaN      NaN      2      25 ...      2      NaN      6
7
top      NaN      NaN  Male      Blouse ...      No      NaN      PayPal
Every 3 Months
freq      NaN      NaN 2652      171 ...      2223      NaN      677
584
mean  1950.500000  44.068462  NaN      NaN ...      NaN      25.351538
NaN      NaN
std  1125.977353  15.207589  NaN      NaN ...      NaN      14.447125
NaN      NaN
min  1.000000  18.000000  NaN      NaN ...      NaN      1.000000
NaN      NaN
25%  975.750000  31.000000  NaN      NaN ...      NaN      13.000000
NaN      NaN
50%  1950.500000  44.000000  NaN      NaN ...      NaN      25.000000
NaN      NaN
75%  2925.250000  57.000000  NaN      NaN ...      NaN      38.000000
NaN      NaN

```

```
max      3900.000000  70.000000  NaN      NaN ...      NaN      50.000000
NaN      NaN
```

```
[11 rows x 18 columns]
```

```
ï»¿Customer ID      0
Age                  0
Gender              0
Item Purchased      0
Category            0
Purchase Amount (USD)  0
Location            0
Size                0
Color               0
Season              0
Review Rating       37
Subscription Status  0
Shipping Type       0
Discount Applied    0
Promo Code Used     0
Previous Purchases  0
Payment Method      0
Frequency of Purchases  0
```

```
dtype: int64
```

```
Data successfully loaded into table 'customer' in database 'customer_behavior'.
```

```
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```

```
C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe
```

```
c:/Users/lra/Downloads/project.py
```

```
ï»¿Customer ID Age Gender Item Purchased Category ... Discount Applied Promo Code
Used Previous Purchases Payment Method Frequency of Purchases
0      1  55 Male Blouse Clothing ... Yes Yes 14
Venmo Fortnightly
1      2  19 Male Sweater Clothing ... Yes Yes 2
Cash Fortnightly
2      3  50 Male Jeans Clothing ... Yes Yes 23 Credit
Card Weekly
3      4  21 Male Sandals Footwear ... Yes Yes 49
PayPal Weekly
4      5  45 Male Blouse Clothing ... Yes Yes 31
PayPal Annually
```

```
[5 rows x 18 columns]
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 3900 entries, 0 to 3899
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```
Data columns (total 18 columns):
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#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
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7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
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13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

	Customer ID	Age	Gender	Item Purchased	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	2	NaN	6	7
top	NaN	NaN	Male	Blouse	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	25.000000	NaN	NaN

75%	2925.250000	57.000000	NaN	NaN ...	NaN	38.000000
NaN	NaN					
max	3900.000000	70.000000	NaN	NaN ...	NaN	50.000000
NaN	NaN					

[11 rows x 18 columns]

```

ĩ»¿Customer ID      0
Age                  0
Gender               0
Item Purchased      0
Category             0
Purchase Amount (USD)  0
Location             0
Size                 0
Color                0
Season               0
Review Rating        37
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0
dtype: int64

```

```

ĩ»¿Customer ID      0
Age                  0
Gender               0
Item Purchased      0
Category             0
Purchase Amount (USD)  0
Location             0
Size                 0
Color                0
Season               0
Review Rating        0
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0
dtype: int64

```

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```
ï»¿Customer ID Age Gender Item Purchased Category ... Discount Applied Promo Code
Used Previous Purchases Payment Method Frequency of Purchases
0      1  55  Male    Blouse Clothing ...      Yes      Yes      14
Venmo      Fortnightly
1      2  19  Male    Sweater Clothing ...      Yes      Yes      2
Cash      Fortnightly
2      3  50  Male    Jeans Clothing ...      Yes      Yes      23  Credit
Card      Weekly
3      4  21  Male    Sandals Footwear ...      Yes      Yes      49
PayPal      Weekly
4      5  45  Male    Blouse Clothing ...      Yes      Yes      31
PayPal      Annually
```

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

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Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	ï»¿Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	str
3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
None

	Customer ID	Age	Gender	Item Purchased	...	Promo Code Used	Previous Purchases
count	3900.000000	3900.000000	3900	3900	...	3900	3900.000000
unique	NaN	NaN	2	25	...	2	NaN
7							6
top	NaN	NaN	Male	Blouse	...	No	NaN
Every 3 Months							PayPal
freq	NaN	NaN	2652	171	...	2223	NaN
584							677
mean	1950.500000	44.068462	NaN	NaN	...	NaN	25.351538
NaN	NaN						
std	1125.977353	15.207589	NaN	NaN	...	NaN	14.447125
NaN	NaN						
min	1.000000	18.000000	NaN	NaN	...	NaN	1.000000
NaN	NaN						
25%	975.750000	31.000000	NaN	NaN	...	NaN	13.000000
NaN	NaN						
50%	1950.500000	44.000000	NaN	NaN	...	NaN	25.000000
NaN	NaN						
75%	2925.250000	57.000000	NaN	NaN	...	NaN	38.000000
NaN	NaN						
max	3900.000000	70.000000	NaN	NaN	...	NaN	50.000000
NaN	NaN						

[11 rows x 18 columns]

Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

```

dtype: int64
ĩ»¿Customer ID      0
Age                 0
Gender              0
Item Purchased      0
Category            0
Purchase Amount (USD)  0
Location            0
Size                0
Color               0
Season              0
Review Rating       0
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0

```

```

dtype: int64
Index(['ĩ»¿customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')

```

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads> &

C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/lra/Downloads/project.py

```

ĩ»¿Customer ID Age Gender Item Purchased Category ... Discount Applied Promo Code
Used Previous Purchases Payment Method Frequency of Purchases
0      1  55 Male   Blouse Clothing ...      Yes      Yes      14
Venmo      Fortnightly
1      2  19 Male   Sweater Clothing ...      Yes      Yes      2
Cash      Fortnightly
2      3  50 Male   Jeans Clothing ...      Yes      Yes      23  Credit
Card      Weekly
3      4  21 Male   Sandals Footwear ...      Yes      Yes      49
PayPal    Weekly
4      5  45 Male   Blouse Clothing ...      Yes      Yes      31
PayPal    Annually

```

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

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3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

	Customer ID	Age	Gender	Item Purchased	...	Promo Code Used	Previous Purchases
count	3900.000000	3900.000000	3900	3900	...	3900	3900.000000
unique	NaN	NaN	2	25	...	2	6
top	NaN	NaN	Male	Blouse	...	No	PayPal
freq	NaN	NaN	2652	171	...	2223	677
mean	1950.500000	44.068462	NaN	NaN	...	NaN	25.351538
std	1125.977353	15.207589	NaN	NaN	...	NaN	14.447125
min	1.000000	18.000000	NaN	NaN	...	NaN	1.000000
25%	975.750000	31.000000	NaN	NaN	...	NaN	13.000000

50%	1950.500000	44.000000	NaN	NaN ...	NaN	25.000000
NaN	NaN					
75%	2925.250000	57.000000	NaN	NaN ...	NaN	38.000000
NaN	NaN					
max	3900.000000	70.000000	NaN	NaN ...	NaN	50.000000
NaN	NaN					

[11 rows x 18 columns]

```

i» Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD)  0
Location           0
Size               0
Color              0
Season             0
Review Rating      37
Subscription Status 0
Shipping Type      0
Discount Applied   0
Promo Code Used    0
Previous Purchases 0
Payment Method     0
Frequency of Purchases 0

```

dtype: int64

```

i» Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD)  0
Location           0
Size               0
Color              0
Season             0
Review Rating      0
Subscription Status 0
Shipping Type      0
Discount Applied   0
Promo Code Used    0
Previous Purchases 0
Payment Method     0

```

Frequency of Purchases 0

dtype: int64

```
Index(['Customer ID', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

age age_group

0 55 Middle-Aged

1 19 Young Adult

2 50 Middle-Aged

3 21 Young Adult

4 45 Middle-Aged

5 46 Middle-Aged

6 63 Senior

7 27 Young Adult

8 26 Young Adult

9 57 Middle-Aged

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads> &

C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/lra/Downloads/project.py

```
Customer ID Age Gender Item Purchased Category ... Discount Applied Promo Code
Used Previous Purchases Payment Method Frequency of Purchases
0 1 55 Male Blouse Clothing ... Yes Yes 14
Venmo Fortnightly
1 2 19 Male Sweater Clothing ... Yes Yes 2
Cash Fortnightly
2 3 50 Male Jeans Clothing ... Yes Yes 23 Credit
Card Weekly
3 4 21 Male Sandals Footwear ... Yes Yes 49
PayPal Weekly
4 5 45 Male Blouse Clothing ... Yes Yes 31
PayPal Annually
```

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

```
# Column Non-Null Count Dtype
---
0 Customer ID 3900 non-null int64
1 Age 3900 non-null int64
```

```

2 Gender          3900 non-null str
3 Item Purchased  3900 non-null str
4 Category        3900 non-null str
5 Purchase Amount (USD) 3900 non-null int64
6 Location        3900 non-null str
7 Size           3900 non-null str
8 Color          3900 non-null str
9 Season         3900 non-null str
10 Review Rating   3863 non-null float64
11 Subscription Status 3900 non-null str
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```

dtypes: float64(1), int64(4), str(13)

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None

	Customer ID	Age	Gender	Item Purchased	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000		
unique	NaN	NaN	2	25	2	NaN	6	
top	NaN	NaN	Male	Blouse	No	NaN	PayPal	
freq	NaN	NaN	2652	171	2223	NaN	677	
mean	1950.500000	44.068462	NaN	NaN	NaN	25.351538		
std	1125.977353	15.207589	NaN	NaN	NaN	14.447125		
min	1.000000	18.000000	NaN	NaN	NaN	1.000000		
25%	975.750000	31.000000	NaN	NaN	NaN	13.000000		
50%	1950.500000	44.000000	NaN	NaN	NaN	25.000000		
75%	2925.250000	57.000000	NaN	NaN	NaN	38.000000		
max	3900.000000	70.000000	NaN	NaN	NaN	50.000000		

[11 rows x 18 columns]

```
ï»¿Customer ID      0
Age                 0
Gender              0
Item Purchased      0
Category            0
Purchase Amount (USD)  0
Location            0
Size                0
Color               0
Season              0
Review Rating       37
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0
dtype: int64
```

```
ï»¿Customer ID      0
Age                 0
Gender              0
Item Purchased      0
Category            0
Purchase Amount (USD)  0
Location            0
Size                0
Color               0
Season              0
Review Rating       0
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0
dtype: int64
```

```
Index(['ï»¿customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
```

```

dtype='str')
age  age_group
0  55  Middle-Aged
1  19  Young Adult
2  50  Middle-Aged
3  21  Young Adult
4  45  Middle-Aged
5  46  Middle-Aged
6  63   Senior
7  27  Young Adult
8  26  Young Adult
9  57  Middle-Aged
purchase_frequency_days frequency_of_purchases

```

```

0      14  Fortnightly
1      14  Fortnightly
2       7   Weekly
3       7   Weekly
4     365   Annually
5       7   Weekly
6      90   Quarterly
7       7   Weekly
8     365   Annually
9      90   Quarterly

```

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads> &

C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/lra/Downloads/project.py

```

İ»¿Customer ID Age Gender Item Purchased Category ... Discount Applied Promo Code
Used Previous Purchases Payment Method Frequency of Purchases
0      1  55  Male   Blouse Clothing ...      Yes      Yes      14
Venmo      Fortnightly
1      2  19  Male   Sweater Clothing ...      Yes      Yes      2
Cash      Fortnightly
2      3  50  Male   Jeans Clothing ...      Yes      Yes      23  Credit
Card      Weekly
3      4  21  Male   Sandals Footwear ...      Yes      Yes      49
PayPal      Weekly
4      5  45  Male   Blouse Clothing ...      Yes      Yes      31
PayPal      Annually

```

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	str
3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

	Customer ID	Age	Gender	Item Purchased	...	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	...	3900	3900.000000		
unique	NaN	NaN	2	25	...	2	NaN	6	
top	NaN	NaN	Male	Blouse	...	No	NaN	PayPal	
freq	NaN	NaN	2652	171	...	2223	NaN	677	
mean	1950.500000	44.068462	NaN	NaN	...	NaN	25.351538		
std	1125.977353	15.207589	NaN	NaN	...	NaN	14.447125		
min	1.000000	18.000000	NaN	NaN	...	NaN	1.000000		
25%	975.750000	31.000000	NaN	NaN	...	NaN	13.000000		
50%	1950.500000	44.000000	NaN	NaN	...	NaN	25.000000		

75%	2925.250000	57.000000	NaN	NaN ...	NaN	38.000000
NaN	NaN					
max	3900.000000	70.000000	NaN	NaN ...	NaN	50.000000
NaN	NaN					

[11 rows x 18 columns]

```

ĩ»¿Customer ID      0
Age                  0
Gender               0
Item Purchased      0
Category             0
Purchase Amount (USD)  0
Location             0
Size                 0
Color                0
Season               0
Review Rating        37
Subscription Status  0
Shipping Type         0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0
dtype: int64

```

```

ĩ»¿Customer ID      0
Age                  0
Gender               0
Item Purchased      0
Category             0
Purchase Amount (USD)  0
Location             0
Size                 0
Color                0
Season               0
Review Rating        0
Subscription Status  0
Shipping Type         0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases  0
dtype: int64

```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

```
age age_group
0 55 Middle-Aged
1 19 Young Adult
2 50 Middle-Aged
3 21 Young Adult
4 45 Middle-Aged
5 46 Middle-Aged
6 63 Senior
7 27 Young Adult
8 26 Young Adult
9 57 Middle-Aged
```

```
purchase_frequency_days frequency_of_purchases
0 14 Fortnightly
1 14 Fortnightly
2 7 Weekly
3 7 Weekly
4 365 Annually
5 7 Weekly
6 90 Quarterly
7 7 Weekly
8 365 Annually
9 90 Quarterly
```

```
<StringArray>
[ 'Fortnightly', 'Weekly', 'Annually', 'Quarterly',
  'Bi-Weekly', 'Monthly', 'Every 3 Months']
```

Length: 7, dtype: str

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads> &

C:\Users\lra\AppData\Local\Python\pythoncore-3.14-64\python.exe

c:/Users/lra/Downloads/project.py

```
ï»¿Customer ID Age Gender Item Purchased Category ... Discount Applied Promo Code
Used Previous Purchases Payment Method Frequency of Purchases
0 1 55 Male Blouse Clothing ... Yes Yes 14
Venmo Fortnightly
1 2 19 Male Sweater Clothing ... Yes Yes 2
Cash Fortnightly
2 3 50 Male Jeans Clothing ... Yes Yes 23 Credit
Card Weekly
```

3	4	21	Male	Sandals	Footwear ...	Yes	Yes	49
PayPal			Weekly					
4	5	45	Male	Blouse	Clothing ...	Yes	Yes	31
PayPal			Annually					

[5 rows x 18 columns]

<class 'pandas.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	str
3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

None

	Customer ID	Age	Gender	Item Purchased	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900 ...	3900	3900.000000		
unique	NaN	NaN	2	25 ...	2	NaN	6	
top	NaN	NaN	Male	Blouse ...	No	NaN	PayPal	
freq	NaN	NaN	2652	171 ...	2223	NaN	677	
mean	1950.500000	44.068462	NaN	NaN ...	NaN	25.351538		
NaN	NaN							

std	1125.977353	15.207589	NaN	NaN ...	NaN	14.447125
NaN	NaN					
min	1.000000	18.000000	NaN	NaN ...	NaN	1.000000
NaN	NaN					
25%	975.750000	31.000000	NaN	NaN ...	NaN	13.000000
NaN	NaN					
50%	1950.500000	44.000000	NaN	NaN ...	NaN	25.000000
NaN	NaN					
75%	2925.250000	57.000000	NaN	NaN ...	NaN	38.000000
NaN	NaN					
max	3900.000000	70.000000	NaN	NaN ...	NaN	50.000000
NaN	NaN					

[11 rows x 18 columns]

```

i»¿Customer ID      0
Age                  0
Gender               0
Item Purchased      0
Category             0
Purchase Amount (USD) 0
Location             0
Size                 0
Color                0
Season               0
Review Rating        37
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases 0
dtype: int64

```

```

i»¿Customer ID      0
Age                  0
Gender               0
Item Purchased      0
Category             0
Purchase Amount (USD) 0
Location             0
Size                 0
Color                0
Season               0
Review Rating        0

```

Subscription Status 0
Shipping Type 0
Discount Applied 0
Promo Code Used 0
Previous Purchases 0
Payment Method 0
Frequency of Purchases 0

dtype: int64

Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount', 'location', 'size', 'color', 'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'promo_code_used', 'previous_purchases',
'payment_method', 'frequency_of_purchases'],
dtype='str')

	age	age_group
0	55	Middle-Aged
1	19	Young Adult
2	50	Middle-Aged
3	21	Young Adult
4	45	Middle-Aged
5	46	Middle-Aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-Aged

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

<StringArray>

['Fortnightly', 'Weekly', 'Annually', 'Quarterly',
'Bi-Weekly', 'Monthly', 'Every 3 Months']

Length: 7, dtype: str

Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount', 'location', 'size', 'color', 'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'previous_purchases', 'payment_method',


```
'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
dtype='str')
```

Data successfully loaded into table 'customer' in database 'customer_behavior'.

PS C:\Users\lra\Downloads>

POSTGRE-SQL (PGADMIN-4)

```
select * from customer limit 20
```

--What is the total revenue generated by male vs female customers

```
select gender, SUM(purchase_amount) as revenue  
from customer  
group by gender
```

-- Which customers used a discount but still spent more than average purchase amount

```
select customer_id, purchase_amount  
from customer  
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from  
customer)
```

-- Which are the top 5 products with the highest average review rating?

```
select item_purchased, ROUND(AVG(review_rating::numeric), 2) as "Average Product Rating"  
from customer  
group by item_purchased  
order by avg(review_rating) desc  
limit 5
```

--Compare the average Purchase Amount between Standard and Express Shipping

```
select shipping_type,  
ROUND(AVG(purchase_amount), 2)  
from customer  
where shipping_type in ('Standard', 'Express')  
group by shipping_type
```

--Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers

```
select subscription_status,  
COUNT(customer_id) as total_customers,  
ROUND(AVG(purchase_amount), 2) as avg_spend,  
ROUND(SUM(purchase_amount), 2) as total_revenue  
from customer
```

```
group by subscription_status
order by total_revenue,avg_spend desc
```

```
--Which 5 products have the highest percentage of purchases with discounts applied?
select item_purchased,
ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT
(*),2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5
```

```
--Segment customers into New,Returning and Loyal based on their total number of previous
purchases and show the count of each segment
with customer_type as (
select i»¿customer_id, previous_purchases,
CASE
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
    END AS customer_segment
from customer
)
select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment
```

```
--What are the top 3 most purchased products within each category
with item_counts as(
select category,
item_purchased,
COUNT(i»¿customer_id) as total_orders,
ROW_NUMBER() over(partition by category order by count(i»¿customer_id) desc) as item_rank
from customer
group by category,item_purchased
)
select item_rank,category,item_purchased,total_orders
from item_counts
where item_rank <= 3
```

```
--Are customers who are repeat buyers (more than 5 previous purchases) also likely to
subscribe?
select subscription_status,
COUNT(i»¿customer_id) as repeat_buyers
```

```
from customer
where previous_purchases > 5
group by subscription_status
```

```
--What is the revenue contribution of each age group
select age_group,
SUM(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc
```